IMAGE-TO-VIDEO MULTIAGENT PIPELINE

AI-Based Automated Image-to-Video Generation System

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Assignment: AI Engineer Technical Assessment

Technologies: Python, FastAPI, React.js, LangGraph, Google Gemini, ChromaDB, Remotion, TypeScript

1. Project Overview

The **Image-to-Video Multiagent Pipeline** is an AI-powered application that automatically transforms a collection of images into a storyboard-based video. Instead of manually arranging images and creating scenes, the system uses multiple AI agents working together to analyze images, understand user requirements, generate a storyboard, create a Remotion video script, and produce the final video output.

The project demonstrates how **Agentic AI**, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, and **LangGraph** can be combined to automate multimedia content creation.

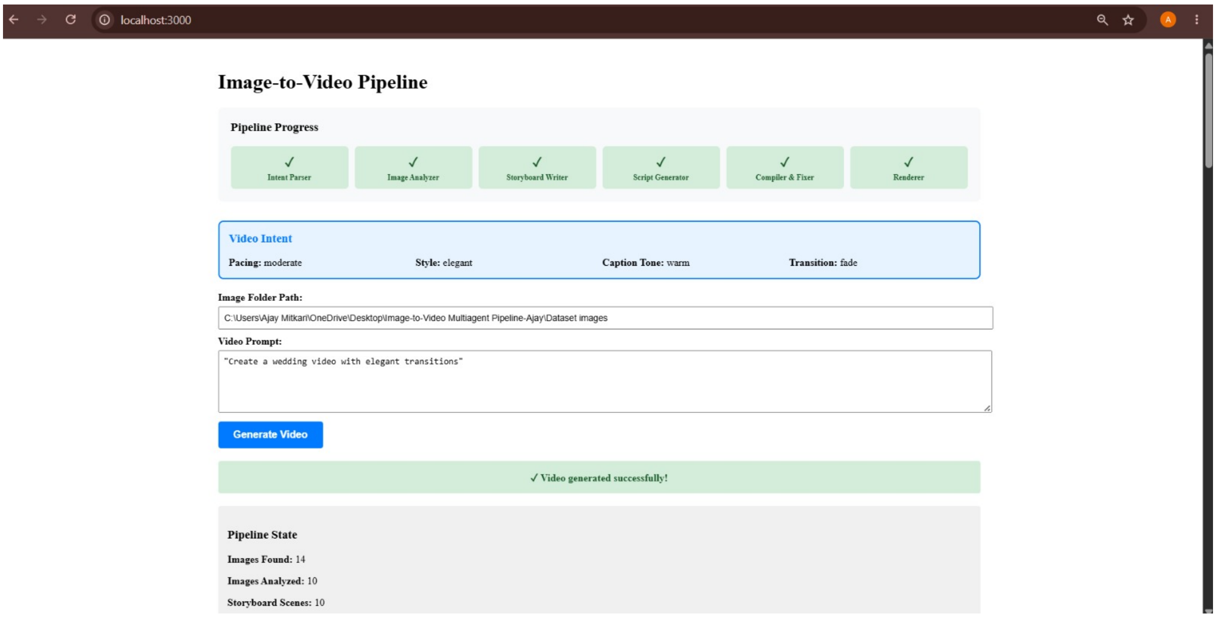

Figure 1: Dashboard & Pipeline Progress

2. Project Objective

The objective of this project is to simplify video creation from image collections by automating the complete workflow using AI agents.

The application performs the following tasks:

- Accepts a folder containing multiple images.
- Accepts a natural language prompt from the user.
- Understands the desired video style.
- Analyzes image content.
- Creates a storyboard.
- Generates a Remotion video script.
- Produces the final MP4 video.

3. Problem Statement

Creating videos manually requires selecting images, arranging scenes, writing captions, adding transitions, and rendering the final output. This process is repetitive and time-consuming.

This project solves the problem by introducing a **Multi-Agent AI Pipeline** where each AI agent performs a specific responsibility, resulting in an automated and efficient video creation process.

4. Technology Stack

Category	Technologies
Programming Language	Python 3.11
Backend	FastAPI
Frontend	React.js, TypeScript, Vite
AI Model	Google Gemini 1.5 Flash
AI Framework	LangGraph, LangChain
RAG Database	ChromaDB
Embedding Model	SentenceTransformers (all-MiniLM-L6-v2)
Video Generation	Remotion
Image Processing	Pillow
Testing	PyTest

5. System Architecture

The application follows a modular architecture where every AI agent performs an independent task.

```
User
   ■
React Frontend
   ■
FastAPI Backend
   ■
LangGraph Pipeline
   ■
Intent Parser
   ■
Image Analyzer
   ■
Storyboard Writer
   ■
Script Generator
   ■
Compiler & Fixer
   ■
```

```
Renderer
   ■
Final MP4 Video
```

The modular architecture improves maintainability, scalability, and debugging.

6. Workflow

The complete workflow consists of six AI agents.

Step 1 – User Input

The user provides:

- Image folder path
- Video generation prompt

Example: *"Create a wedding video with elegant transitions and warm captions."*

The application sends the request to the FastAPI backend.

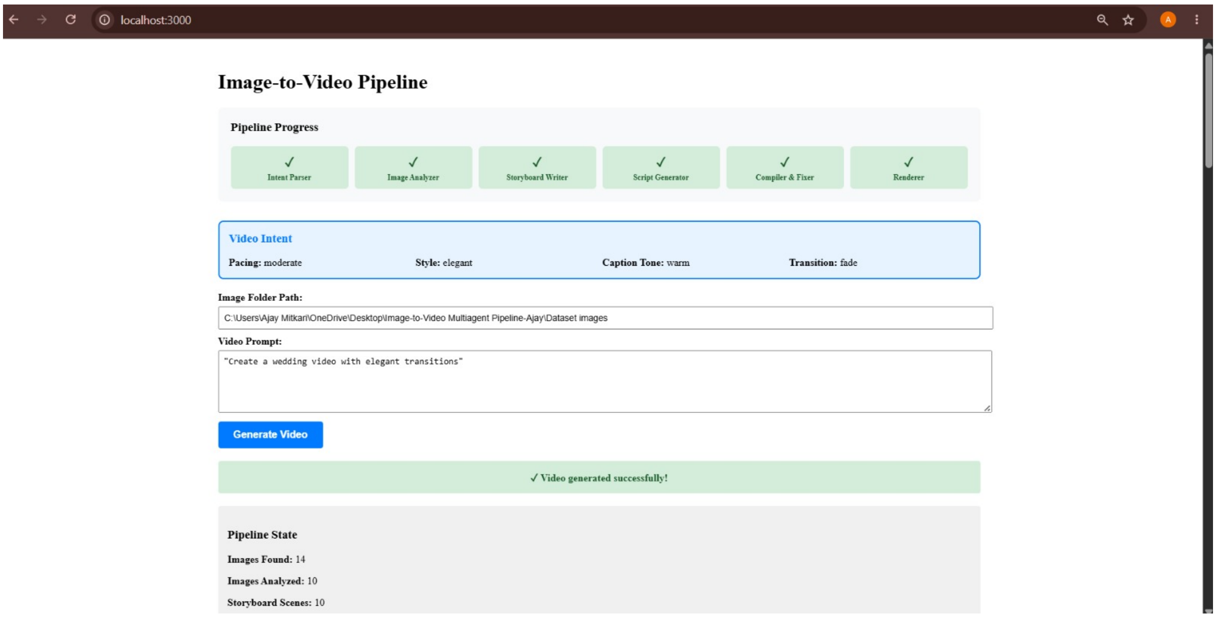

Figure 1 (repeated): User Input and Pipeline Progress

Step 2 – Intent Parser

The Intent Parser uses Google Gemini to understand the user's prompt.

It extracts important information such as:

- Video Style
- Caption Tone
- Transition Type
- Video Pacing

Example Output:

- Style: Elegant
- Transition: Fade
- Caption Tone: Warm
- Pacing: Moderate

Step 3 – Image Analyzer

The Image Analyzer processes all images inside the selected folder.

The system identifies:

- Image quality
- Scene information
- Number of people
- Overall description

Only the best images are selected for storyboard generation.

Pipeline State Example:

- Images Found: 14
- Images Analyzed: 10

Step 4 – Storyboard Writer

The Storyboard Writer uses image analysis together with RAG-based style guides stored in ChromaDB.

It automatically generates scenes.

Each scene contains:

- Image
- Caption
- Duration
- Transition

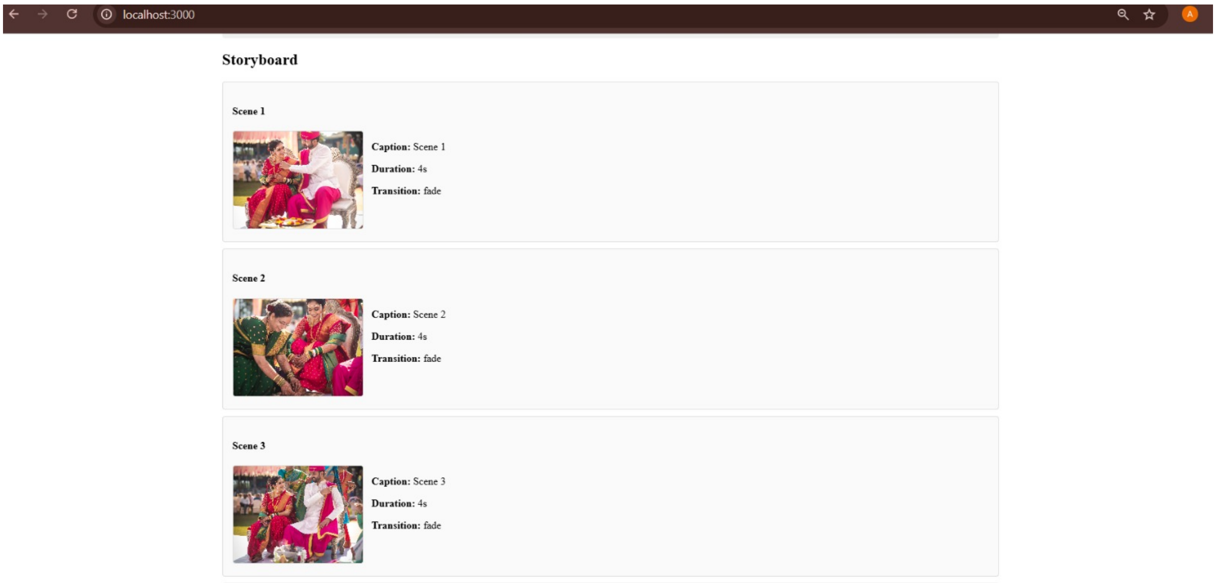

Figure 2: Storyboard Generation (Scenes 1–3)

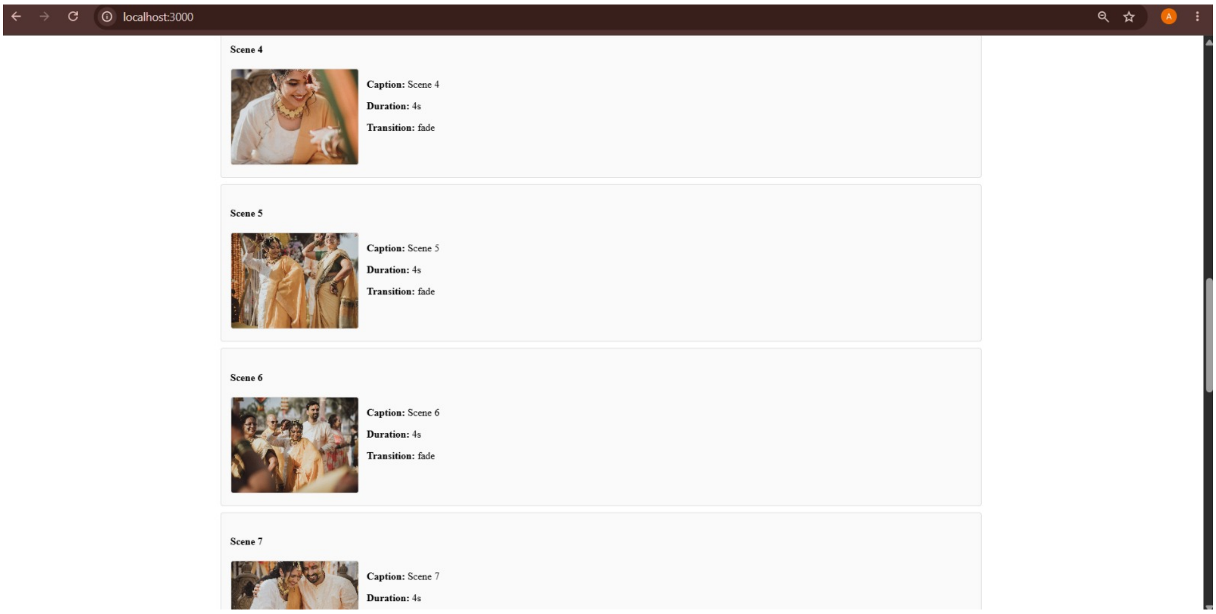

Figure 3: Storyboard Continuation (Scenes 4–7)

Step 5 – Script Generator

The Script Generator converts the storyboard into a complete **Remotion TypeScript** script.

The generated script includes:

- React components
- Image sequences
- Captions
- Timing
- Transitions

The system retrieves Remotion documentation from ChromaDB to improve script quality.

Step 6 – Compiler & Fixer

Before rendering, the generated script is validated.

The Compiler & Fixer:

- Detects script issues
- Automatically retries up to three times
- Produces a corrected script if necessary

This improves the reliability of the pipeline.

Step 7 – Renderer

The Renderer creates the final video output.

The generated video is saved as:

```
sample_output/final.mp4
```

The final video is also displayed inside the application.

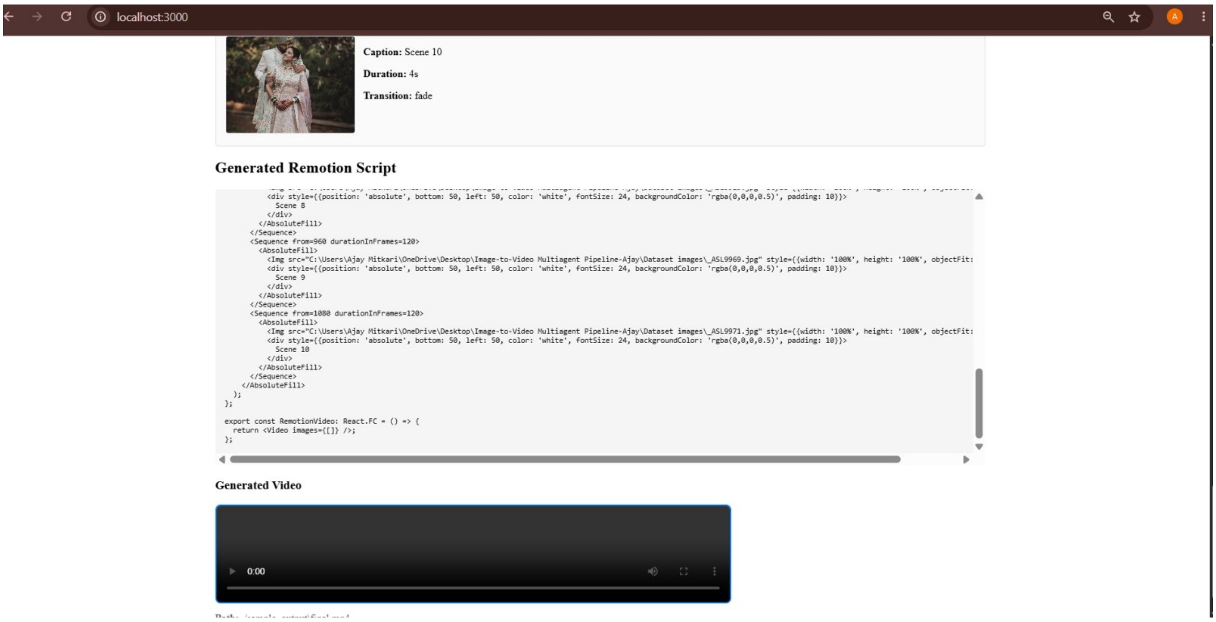

Figure 4: Generated Remotion Script & Final Video Output

7. User Interface

The web application provides a clean and interactive interface.

The UI contains:

- Image Folder Selection
- Video Prompt Input
- Generate Video Button
- Pipeline Progress
- Pipeline State

- Storyboard Preview
- Generated Remotion Script
- Embedded Video Player

The application allows users to monitor every stage of the pipeline.

8. Pipeline Progress

The application visually displays the execution status of every AI agent.

Completed stages include:

- ✔ Intent Parser
- ✔ Image Analyzer
- ✔ Storyboard Writer
- ✔ Script Generator
- ✔ Compiler & Fixer
- ✔ Renderer

This provides real-time feedback to the user during execution.

9. Retrieval-Augmented Generation (RAG)

The project integrates a lightweight RAG system using ChromaDB.

Two collections are maintained:

Style Guide Collection

Contains recommendations for:

- Wedding Videos
- Birthday Videos
- Corporate Videos
- Travel Videos

These guides help generate better storyboards.

Remotion Documentation Collection

Stores documentation related to:

- Sequence
- Img
- AbsoluteFill
- Animation Components

This information helps generate accurate Remotion scripts.

10. Project Features

The project includes the following features:

- Multi-Agent AI Architecture
- Prompt Understanding using Gemini
- AI-based Image Analysis
- Automatic Storyboard Creation
- RAG Integration with ChromaDB
- Dynamic Remotion Script Generation
- Automatic Script Validation
- Video Rendering
- Interactive React Dashboard
- Video Preview
- REST API Architecture

11. Project Output

The successful execution of the pipeline produces:

- AI-generated storyboard
- Scene thumbnails
- Video intent summary
- Pipeline execution details
- Generated Remotion TypeScript script
- Final MP4 video

The screenshots below demonstrate the successful execution of the complete workflow.

- **Figure 1:** Dashboard & Pipeline Progress
- **Figure 2:** Storyboard Generation
- **Figure 3:** Storyboard Continuation
- **Figure 4:** Generated Remotion Script & Final Video Output

12. Challenges Faced

During development, the following technical challenges were addressed:

- Prompt interpretation using LLMs
- Image quality filtering
- Storyboard sequencing
- AI response validation
- Remotion script generation

- Backend–Frontend integration
- Video rendering workflow
- RAG knowledge retrieval

13. Future Enhancements

Future improvements can include:

- AI-generated voice narration
- Background music generation
- Automatic subtitle generation
- Cloud deployment
- User authentication
- Multiple video themes
- Social media export formats
- Support for multiple languages

14. Conclusion

The **Image-to-Video Multiagent Pipeline** successfully demonstrates the practical use of Agentic AI, LangGraph, Retrieval-Augmented Generation (RAG), and Large Language Models to automate video creation from image collections.

The modular multi-agent architecture enables the system to understand user intent, analyze images, generate a storyboard, produce a Remotion video script, validate the generated output, and render the final video in a structured and scalable manner.

This project showcases modern AI engineering concepts including workflow orchestration, multimodal AI, prompt engineering, RAG, and full-stack application development. It demonstrates the ability to integrate AI models with backend services and frontend interfaces to build an end-to-end production-style application.

Appendix

Screenshots included in this report, in order:

- **Figure 1:** Dashboard & Pipeline Progress
- **Figure 2:** Storyboard Generation
- **Figure 3:** Storyboard Continuation
- **Figure 4:** Generated Remotion Script & Final Video Output